Electric Vehicle Market Performance & Insights

SkillBridge Course Project — Project Documentation

Project Title: Electric Vehicle Market Performance & Insights

Project Type: SkillBridge course project

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Programme: B.Tech — CSE (AI & ML)

Semester: 3rd Semester

Section: IBM-1(A)

University: Medicaps University, Indore

Faculty / Project Guide: _____

Academic Session: _____

Submission Date: _____

Add the guide name, academic session, and submission date if required by the university.

Declaration

I declare that this project documentation titled “Electric Vehicle Market Performance & Insights” describes the work carried out as part of the SkillBridge course project. The report documents the dataset, data preparation, Tableau analysis, dashboard and Story, and web integration used for the project.

Student Signature: _____ Date: _____

Acknowledgement

I would like to thank Medicaps University and the SkillBridge course programme for the opportunity to work on this data analytics project. I also acknowledge the Electric Vehicle Market Classification Dataset shared through SkillWallet and hosted on Kaggle. This project helped me practise data preparation, visual analytics, dashboard design, and web integration using Tableau Public, Python Flask, HTML, and CSS.

Abstract

Electric Vehicle Market Performance & Insights is a data analytics project that explores electric-vehicle attributes and market indicators through interactive visualizations. The project uses training and testing CSV files from the Electric Vehicle Market Classification Dataset on Kaggle, accessed through a link provided in SkillWallet. The training dataset contains 200,000 rows and 37 columns, while the testing dataset contains 100,000 rows and 37 columns. The dataset includes vehicle identity, manufacturer, model year, manufacturing country, market region, battery specifications, range, efficiency, motor and charging characteristics, pricing, safety, customer ratings, annual sales, and estimated CO■ savings.

The data was prepared for analysis and visualized in Tableau Public. The workbook contains KPI sheets, a dashboard titled “EV Market Performance & Insights,” and a Story titled “Electric Vehicle Market Insights.” The dashboard displays average battery capacity of 78.24 kWh, 200 manufacturers, average customer rating of 4.089, 199,999 EV records, and 17,980 EV models. It includes comparisons by manufacturer and country and trends by model year and annual sales. A web interface built with Flask, HTML, and CSS is used to present the project and integrate the published Tableau visualization where embedding is supported.

The purpose of the project is to make a large dataset easier to explore and explain. Dashboard values are dataset-derived and should be interpreted according to the fields and aggregation methods used. The source describes the dataset as synthetic and designed for classification practice, so the results should not be treated as verified real-world EV market statistics.

1. Introduction

1.1 Background

Electric vehicles are an important area of automotive technology and market analysis. EV datasets may include battery capacity, range, efficiency, pricing, manufacturer, country, model year, sales, customer rating, and environmental measures. Organizing these attributes into visual summaries can help users explore patterns more efficiently than reading raw rows.

1.2 Problem Statement

The project addresses the difficulty of understanding a large, multi-column EV dataset through raw tables alone. It creates a Tableau dashboard and Story to summarize key measures and compare vehicle records across manufacturers and manufacturing countries, while showing model-year and annual-sales trends.

1.3 Objectives

• Inspect the training and testing CSV files supplied for the project.

• Prepare the dataset for visualization and analysis.

• Create KPI views for selected EV measures.

• Compare vehicle records by manufacturer and manufacturing country.

• Visualize vehicle-record counts by model year and annual sales by year.

• Organize the visualizations into a Tableau Story.

• Build a Flask-based web interface using HTML and CSS.

• Document implementation, testing, limitations, and future improvements.

1.4 Scope

This project focuses on descriptive analytics and dashboard communication. It does not claim to train or evaluate a predictive classification model. The Kaggle dataset is described by its publisher as synthetic, so its patterns are useful for analytical practice rather than direct evidence of actual market share or real-world EV sales.

2. Tools and Technologies

• Tableau Public — worksheets, KPI views, dashboard composition, filters, and Story points.

• Python — programming language used by the Flask application.

• Flask — serves the project website locally.

• HTML — defines the page structure and content.

• CSS — styles the website layout and appearance.

• CSV — format of the training and testing datasets.

• Visual Studio Code — editing and local development environment.

• GitHub — source-code repository.

Local development address: http://127.0.0.1:5000/ . This address is accessible on the development machine and is not, by itself, a public deployment URL.

3. Data Collection and Extraction

3.1 Dataset Source

Dataset title: Electric Vehicle Market Classification Dataset. Creator shown on Kaggle: Harish Reddy. Source URL: https://www.kaggle.com/datasets/harishreddy22/electric-vehicle-market-classification-dataset

The dataset was downloaded from Kaggle using a link provided through SkillWallet. The Kaggle dataset description identifies the data as synthetic and intended for multi-class classification practice across 15 market segments. Rows are procedurally generated, so the dataset should be treated as a practice dataset rather than an authoritative real-world market census.

3.2 Files and Dimensions

File	Rows	Columns	Role
ev_market_train.csv	200,000	37	Training dataset supplied with the project
ev_market_test.csv	100,000	37	Testing dataset supplied with the project

3.3 Main Attributes

The fields include vehicle_id, model_name, manufacturer, model_year, manufacturing_country, market_region, category, battery_capacity_kwh, battery_type, battery_cycle_life, battery_degradation_pct_5yr, range_km, efficiency_wh_per_km, motor_power_kw, torque_nm, num_motors, drivetrain, top_speed_kmph, acceleration_0_100_sec, max_dc_charging_kw, charging_dc_fast_time_min, charging_ac_time_hr, num_charging_ports, num_seats, weight_kg, cargo_capacity_liters, price_usd, insurance_cost_usd_yearly, warranty_years, resale_value_pct_3yr, safety_rating_stars, autonomous_level, infotainment_screen_inch, co2_saved_tons_per_year, annual_sales_units, customer_rating, and recommended_use_case.

3.4 Collection Procedure

1. Open the Kaggle dataset link supplied through SkillWallet.

2. Download the training and testing CSV files.

3. Save the files in the project workspace.

4. Open Tableau Public and connect to the required CSV file through the Data Source page.

5. Review field names and data types before building worksheets.

4. Data Preparation

The student confirmed that the data was cleaned. The exact operations used were not fully specified in the available notes, so the report does not claim particular replacements or deletions that have not been confirmed.

4.1 Preparation Activities to Document

• Inspect field names and data types.

• Check null or blank values in relevant columns.

• Check whether vehicle_id is unique and how null IDs affect record counts.

• Check numeric fields for invalid or unexpected values.

• Review category labels for inconsistent spelling or whitespace.

• Apply only justified cleaning operations and record each one.

• Recheck Tableau aggregations and KPI values after preparation.

4.2 Data Quality Note

An initial inspection identified missing values in columns such as battery_cycle_life, resale_value_pct_3yr, infotainment_screen_inch, annual_sales_units, and customer_rating. The exact treatment of those missing values should be described according to the operations actually performed during cleaning.

NOTE: The Tableau dashboard shows 199,999 EV records while the training file has 200,000 rows. Verify whether the KPI counts non-null Vehicle Id values or uses a distinct count before describing it as the total number of source rows.

5. Data Visualization and Analysis

5.1 KPI Cards

KPI	Displayed value	Meaning
Average Battery Capacity	78.24 kWh	Average battery capacity according to the worksheet aggregation
Total Manufacturers	200	Manufacturer count shown in the dashboard
Average Customer Rating	4.089	Mean customer-rating value shown
Total EV Records	199,999	Record/vehicle-ID measure shown; aggregation should be verified
Total EV Models	17,980	Model count shown in the dashboard

5.2 Charts

• Electrical Vehicle Records by Manufacturer — horizontal bars comparing distinct Vehicle Id counts by manufacturer; a Top 10 limit is visible.

• EVs by Country — horizontal bars comparing distinct Vehicle Id counts by manufacturing country.

• EVs by Model Year — line chart showing distinct Vehicle Id counts by model year.

• Annual EV Sales — line chart showing Annual Sales Units by model year.

5.3 Filters

The dashboard screenshot shows filters for Manufacturer and Manufacturing Country. The manufacturer visualization also has a Top 10 limit based on COUNTD([Vehicle Id]). Filters allow users to narrow the displayed data.

5.4 Interpretation

The manufacturer and country charts show counts of distinct vehicle IDs; they do not directly show annual sales. The Annual EV Sales chart uses annual sales units. These measures should not be treated as interchangeable. Trends in a synthetic dataset are examples of patterns in the generated data and are not evidence of actual global EV market changes.

6. Dashboard Design

Dashboard title: EV Market Performance & Insights.

The dashboard places KPI cards along the top. Manufacturer and country comparison charts appear in the middle, with model-year and annual-sales line charts beneath them. Filter controls are placed along the right side. The visual design uses pale teal KPI headers, blue chart marks, and a light background.

The dashboard is intended to provide a consolidated view of selected EV measures and allow interactive exploration using the manufacturer and manufacturing-country filters.

7. Tableau Story and Insights

Story title: Electric Vehicle Market Insights.

7.1 Story Points

• An interactive analysis of electric vehicles — overview of the dashboard.

• Manufacturer Analysis — comparison of records across manufacturers.

• Geographical Analysis — comparison of records across manufacturing countries.

• Model Year Trends — model-year view of vehicle-record counts.

• Annual Sales Trends — annual sales view by model year.

7.2 Findings Supported by the Screenshots

• The dashboard displays 200 manufacturers and 17,980 EV models.

• The manufacturer chart ranks manufacturers by distinct Vehicle Id count and applies a Top 10 limit.

• The country chart compares distinct vehicle-ID counts across the countries included in the dataset.

• The annual-sales line shows variation by year and a visible peak around 2024 in the supplied screenshot; exact values should be read from Tableau before quoting them.

These are descriptive observations of the supplied synthetic dataset, not claims about verified market conditions.

8. Web Integration

8.1 Architecture

Flask serves the website, HTML provides the page structure, and CSS controls its appearance. The Tableau Public view is intended to be displayed in an iframe after publishing the view and obtaining the correct embed URL.

8.2 Project Structure

```
EV_Insights/\n■■■ app.py\n■■■ index.html\n■■■ static/\n    ■■■ style.css
```

8.3 Local Run Procedure

1. Open the project folder in Visual Studio Code.

2. Ensure Python and Flask are installed in the active environment.

3. Run the application with the command configured for app.py (commonly `python app.py`).

4. Open http://127.0.0.1:5000/ in the browser.

5. Verify that the HTML page and CSS load correctly.

6. Open the published Tableau view directly, then check whether the embedded view appears on the page.

8.4 Tableau Embed Considerations

The Tableau view must be published and accessible for the iframe to load. Use Tableau Public's Share or Embed option to obtain the correct URL. If the iframe refuses to connect, verify the published view URL and the public accessibility settings.

8.5 Repository

GitHub repository: https://github.com/prakalpjain025-max/EV-Market-Performance-Insights

The GitHub repository provides source-code access. A repository URL is not necessarily a live deployed website URL.

9. Testing and Validation

The following cases should be executed and their actual outcomes recorded. The document does not assume that every test passed.

ID	Test	Expected result	Actual result
TC-01	Open Flask site locally	Page loads without a server error	To be recorded
TC-02	Load CSS	Styling is applied	To be recorded
TC-03	Open Tableau view directly	Published view loads	To be recorded
TC-04	Open embedded dashboard	Dashboard appears in the web page	To be recorded
TC-05	Change Manufacturer filter	Charts respond as configured	To be recorded
TC-06	Change Manufacturing Country filter	Country selection filters the dashboard	To be recorded
TC-07	Navigate Story points	Each selected story point displays	To be recorded
TC-08	Validate KPI calculations	Cards match their worksheet calculations	To be recorded
TC-09	Inspect labels and axes	Titles and units are readable	To be recorded
TC-10	Open GitHub repository	Project files are accessible	To be recorded

Tableau verification was confirmed by the student. The actual outcomes of individual web-embed and browser tests should be recorded based on the latest run.

10. Limitations and Future Improvements

10.1 Limitations

• The dataset is synthetic and intended for classification practice, not a verified source of real-world EV market statistics.

• The exact cleaning steps should be documented from the actual workflow.

• KPI values depend on aggregation, filters, and null handling.

• The Tableau embed depends on the published view being accessible.

• The local Flask address is not a public deployment URL.

10.2 Future Improvements

• Add source attribution and a dataset refresh/version note.

• Explain KPI definitions and aggregation methods in tooltips.

• Add views for range, efficiency, price, battery degradation, and CO■ savings if relevant.

• Test responsive layout and readability on mobile devices.

• Deploy the Flask website if a public demo is required.

• Add repeatable data-quality checks for missing values, duplicate IDs, and KPI consistency.

11. Conclusion

Electric Vehicle Market Performance & Insights demonstrates a workflow for exploring a structured EV dataset and communicating results through Tableau Public. The dashboard summarizes selected measures and compares vehicle records by manufacturer and manufacturing country, while line charts show model-year and annual-sales patterns. The Tableau Story organizes the analysis into a guided sequence. A Flask-based website using HTML and CSS provides a web interface for presenting the project and embedding the published dashboard. The dataset's synthetic nature should be stated clearly, and the final submission should include the exact cleaning steps, verified KPI definitions, and actual testing outcomes.

Appendix A — User Guide

1. Open the published dashboard or local Flask project page.

2. Review KPI cards at the top of the dashboard.

3. Use the Manufacturer filter to focus on selected manufacturers.

4. Use the Manufacturing Country filter to focus on selected countries.

5. Interpret manufacturer and country bars as distinct vehicle-ID counts.

6. Review model-year counts and annual sales in their respective line charts.

7. Use the Story navigation to move through the overview, manufacturer, geographical, model-year, and sales views.

8. Hover over marks to read tooltips if the workbook has them configured.

Appendix B — GitHub README Draft

```
# Electric Vehicle Market Performance & Insights\n\nA SkillBridge data analytics course project exploring elec
tric-vehicle attributes with Tableau Public and a Flask-based web interface.\n\n## Student\n- Name: Prakalp Ja
in\n- Roll number: EN25CS3090152\n- Programme: B.Tech CSE (AI & ML), 3rd Semester, Section IBM-1(A)\n\n## Tool
s\n- Tableau Public\n- Python / Flask\n- HTML and CSS\n- CSV datasets\n\n## Dataset\nThe project uses `ev_mark
et_train.csv` (200,000 rows, 37 columns) and `ev_market_test.csv` (100,000 rows, 37 columns) from the [Electri
c Vehicle Market Classification Dataset on Kaggle](https://www.kaggle.com/datasets/harishreddy22/electric-vehi
cle-market-classification-dataset). The Kaggle publisher describes the dataset as synthetic.\n\n## Dashboard\n
The dashboard includes KPI cards for average battery capacity, manufacturer count, customer rating, EV record
count, and EV model count. Visualizations compare vehicle records by manufacturer and country and show model-y
ear and annual-sales trends. The Tableau Story covers overview, manufacturer analysis, geographical analysis,
model-year trends, and annual-sales trends.\n\n## Run locally\n1. Install Python and Flask.\n2. Open the proje
ct folder in a terminal.\n3. Run `python app.py` if that is the command configured in your application.\n4. Vi
sit `http://127.0.0.1:5000/`.\n\n## Repository\nhttps://github.com/prakalpjain025-max/EV-Market-Performance-In
sights\n\n## Notes\nThe local Flask URL is not a public deployment URL. The Tableau view must be published and
 accessible for iframe embedding. Verify KPI calculations and complete actual test results before final submis
sion.
```

Appendix C — Final Submission Checklist

■ Confirm faculty/guide name, academic session, and submission date.

■ Keep the Kaggle source URL and synthetic-data caveat in the report.

■ Record exact cleaning steps actually performed.

■ Verify the Total EV Records calculation against the data and Tableau worksheet.

■ Insert screenshots of the final dashboard, Story, and web page.

■ Complete actual test results.

■ Confirm the published Tableau URL and test the iframe.

■ Follow any official SkillBridge template if one becomes available.